

Certificate of Analysis

PRODUCT DETAILS

Product Name	Exosomes (RejuvCORE*)
Batch	
Manufacturing Date	
Expiry Date	
Dosage Form	Clear-colourless solution
Storage Temperature	4 – 8°C

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	
Sterility	No cloudiness of the test media	Negative	
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	
Mycoplasma	Mean of absorbance <0.05	Negative	
Simple Western Blot	Positive Marker: Tetraspanin – CD9, CD63 Cytosolic – TSG101 Negative Marker: Cellular – Calnexin	Detected Trace marker detected	
Nanoparticle Tracking Analysis (NTA)	Exosome particle size range 30nm – 200nm	34 x 10 ⁹ particles/mL	

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Storage recommendation:

- 1) Refrigerate the vials in 4-8°C at all times upon received
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolong exposure in room condition
- 4) It is recommended to place the vials in freezer for prolonged storage

*General Exosomes fortified with Curcumin

Authorised Signature:

Quality Control Department

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Supporting Information

STERILITY TEST ^{6,8}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST ^{3,6}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST ^{4,6}

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

SIMPLE WESTERN BLOT

Marker	Positive Marker		Negative Marker	
	Tetraspanin		Cytosolic	Cellular
	CD9	CD63	TSG101	Calnexin
Result	+	+	+	*

'+' indicates the presence of the marker in the cell.
'*' indicates the trace of the marker detected in the cell

NANOPARTICLE TRACKING ANALYSIS (NTA) ^{7,8}

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.



REFERENCES

1. Chetthimada, Sukrutha, Lorenz, David R., Misra, Vikas, Dillon, Simon T., Reeves, R. Keith, Manickam, Cordelia, 2018, Exosome markers associated with immune activation and oxidative stress in HIV patients on antiretroviral therapy.
2. Joana Maia, Silvia Batista, Nuno Couto, Ana C. Gregório, Cristian Bodo, Julia Etzanowska, Maria Carolina Strano Moraes, Bruno Costa-Silva, 2020, Employing Flow Cytometry to Extracellular Vesicles Sample Microvolume Analysis and Quality Control.
3. Bacterial Endotoxins. European Pharmacopoeia 5.0; <https://empua.com/validation/Method/LAL/EUPHARMACPOEIA.pdf>
4. US Pharmacopeia <83> Mycoplasma Tests
5. CH Harmonised Tripartite Guideline. Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q5B.
6. US Pharmacopeia <71> Sterility Test
7. Neha Chopra, Ibrahim Dutt Aray, Namrata Jain, Poonam Yadav, Salma Wajid, Surinder P. Singh, Sangeta Choudhary, 2019, Biophysical Characterization and Drug Delivery Potential of Exosomes from Human Wharton's Jelly-Derived Mesenchymal Stem Cells.
8. Dae Hyun Ha, Hyun-keun Kim, Joon Lee, Hyuck-Hoon Kwon, Gyeong-Hun Park, Seon-Ho Seong Yang, Jee Yoon Jung, Housung Choi, Jun Ho Lee, Sumi Sung, Yong Weon Yi, Byoung Seung Cho, 2020, Mesenchymal Stem/Stromal Cell-Derived Exosomes for Immunomodulatory Therapeutics and Skin Regeneration.